

# Low loss Digitally Tunable Dispersion Controller Using Ridge Waveguides

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**Abstract** We demonstrate a low-loss digitally tunable dispersion controller based on rib waveguides on a silicon-on-insulator (SOI) platform. The device achieves a digital dispersion tuning range of  $-251.87$  to  $+262.33$  ps/nm over the  $1544$ – $1553.2$  nm wavelength band. ©2026 The Author(s)

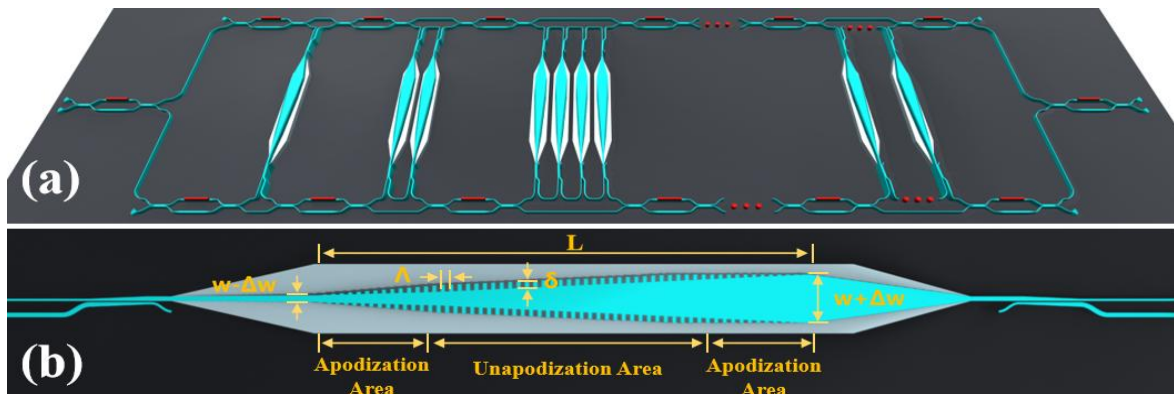
## 1. Introduction

Driven by the increasing demands of high-capacity optical communications [1] and broadband microwave photonics (MWP) [2][3], on-chip dispersion management has emerged as a critical research frontier. Over the past few years, various on-chip dispersion controllers have been developed using architectures such as cascaded microring resonators (MRRs) [4], cascaded Mach-Zehnder interferometers (MZIs) [5], and waveguide Bragg gratings [6]. However, MZI-based devices are generally limited by large footprint size and relatively limited bandwidth. MRRs offer compact footprints but require precise and complex thermal tuning to stabilize their spectral responses. In contrast, Bragg-grating-based devices provide broadband, engineerable dispersion characteristics with continuous spectral control. At the system level, digitally tunable dispersion controllers (DTDCs) have been demonstrated by cascading these elementary structures to realize discrete, reconfigurable delay states [7][8].

Beyond architectural design, material platform selection also plays a crucial role in determining device performance. Silicon-on-insulator (SOI) strip waveguides enable sub-micron bending radii ( $\sim 5$   $\mu\text{m}$ ), supporting high-density integration. However, the ultra-high index contrast leads to increased propagation loss and strong sensitivity to fabrication-induced variations. For previous

DTDCs on SOI, the propagation loss typically exceeded 10 dB/ns [7][8]. By comparison, silicon nitride (SiN) platforms exhibit significantly lower intrinsic loss [6] but require much larger bending radii ( $>50$   $\mu\text{m}$ ), resulting in increased device footprints [9]. Additionally, unlike SOI, SiN platforms face challenges in seamlessly integrating active components, such as high-speed modulators and photodetectors, due to material compatibility constraints.

In this work, we propose and demonstrate a reconfigurable low loss DTDC capable of providing both positive and negative dispersion by incorporating rib-waveguide bidirectional chirped multimode waveguide gratings (CMWGs) on a SOI platform. To simultaneously optimize optical performance and integration density, a hybrid waveguide integration strategy is adopted: rib waveguides are employed within the grating sections to reduce scattering loss and enable precise modal control, while strip waveguides are used for compact routing and interconnections. This approach exploits the tight bending capability of strip waveguides to minimize chip footprint without compromising the overall power budget. The proposed device is experimentally realized with a footprint of  $4.85 \times 0.4$  mm<sup>2</sup> and achieves a digitally reconfigurable dispersion range from  $-251.87$  to  $+262.33$  ps/nm over the  $1544$ – $1553.2$  nm band, with low losses of 1.67 dB/ns for positive dispersion and 1.40 dB/ns for negative dispersion.



**Fig.1:** (a) Configurations of the proposed low loss DTDC; (b) Structural details of the integrated CMWG .

## 2. Structure and design

Figure 1(a) shows the architecture of the proposed DTDC, which comprises  $N$  stages of bidirectional chirped multimode waveguide gratings (CMWGs) and thermo-optic  $2 \times 2$  Mach-Zehnder switches (MZSs). All the bidirectional CMWG devices have adopted the same design for ease of expansion, and they all have the same dispersion  $D_0$ . A distinctive advantage of the CMWG is its multimode operation, which facilitates the spatial separation of reflected signals from the excitation port, effectively eliminating the requirement for on-chip optical circulators. Beyond this, the CMWGs allow the dispersion sign to be determined by the light's propagation direction. To enable digital tunability, the number of CMWGs in the  $n^{\text{th}}$  stage scales as  $m = 2^{n-1}$ , providing the discrete delay states necessary for precise dispersion control.

By coordinating the input and output switches to reverse the optical path through the CMWG sections, the global dispersion polarity of the entire DTDC can be selectively toggled between positive and negative regimes, achieving reconfigurability. Therefore, the total dispersion of the current positive/negative dispersion compensator is accumulated by all levels and can be digitally adjusted by controlling the states of all  $2 \times 2$  MZSs. With  $(2^N - 1)$  bidirectional units, the DTDC allows the number of active units  $P$  to be selected within the range of 0 to  $(2^N - 1)$ . This enables reconfigurable delay states spanning from  $-(2^N - 1)D_0$  and  $(2^N - 1)D_0$ , in steps of  $D_0$ .

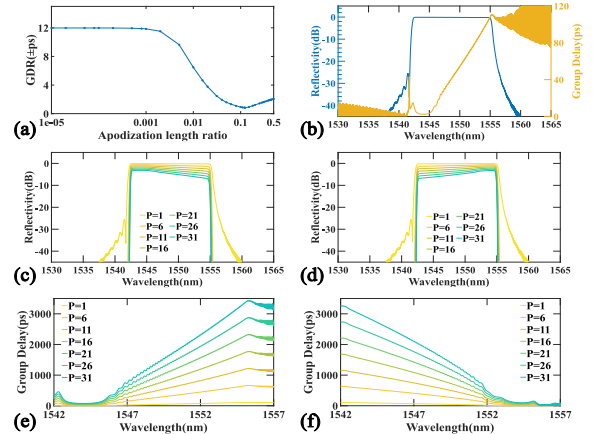
$$\Lambda = \lambda(n_{\text{eff}_{TE0}} + n_{\text{eff}_{TE1}}) \quad (1)$$

Figure 1(b) shows the structure of the key component CMWG for dispersion control. Based on coupled-mode theory, the forward  $TE_0$  mode is coupled to the backward  $TE_1$  mode under the condition specified in Eq. (1). The grating chirp is implemented by linearly tapering the waveguide width from  $w_0 - \Delta w$  to  $w_0 + \Delta w$  at a constant period  $\Lambda$ . We chose  $w_0 = 1.5 \mu\text{m}$  to minimize sidewall scattering loss by reducing the modal field overlap with the waveguide edges, resulting in a calculated period of  $\Lambda = 280 \text{ nm}$  for  $1550 \text{ nm}$  operation. To balance fabrication constraints and the suppression of group delay ripple (GDR), a small corrugation depth  $\delta_0$  is utilized, with a maximum value of  $150 \text{ nm}$ . The grating length  $L$  is fixed at  $4 \text{ mm}$  to provide the target total delay with a duty cycle of  $0.5$ . Since there is a trade-off between dispersion and bandwidth for a fixed  $L$ , we optimized  $\Delta w = 300 \text{ nm}$  to maintain significant dispersion across the required spectral range. Moreover, to suppress parasitic Fabry-Perot resonances and the subsequent GDR, the corruga-

tion depth  $\delta(z)$  is apodized at both terminals according to the following modulation function:

$$\delta(z) = \begin{cases} \delta_0 \sin\left(\frac{\pi z}{2\beta L}\right), & 0 \leq z \leq \beta L \\ \delta_0, & \beta L < z \leq (1 - \beta)L \\ \delta_0 \sin\left(\frac{\pi(L - z)}{2\beta L}\right), & (1 - \beta)L < z \leq L \end{cases} \quad (2)$$

where  $\beta$  denotes the apodization length ratio relative to the total grating length  $L$ . Here, the transmission matrix method is used to conduct simulation calculations for the device. As shown in Fig 2(a), we simulated the GDR values calculated when  $\beta$  increased from 0 to  $1/2$ . Here, we select  $\beta = 0.1$  to obtain a high linearity and low jitter group delay spectrum. At this time, the designed bidirectional CMWG corresponds to a  $3 \text{ dB}$  bandwidth of approximately  $13 \text{ nm}$ , a maximum group delay of  $110 \text{ ps}$ , and a dispersion of approximately  $10.01 \text{ ps/nm}$ , as shown in fig2(b). When the light is switched from the other end, the corresponding negative dispersion is  $-9.44 \text{ ps/nm}$ .

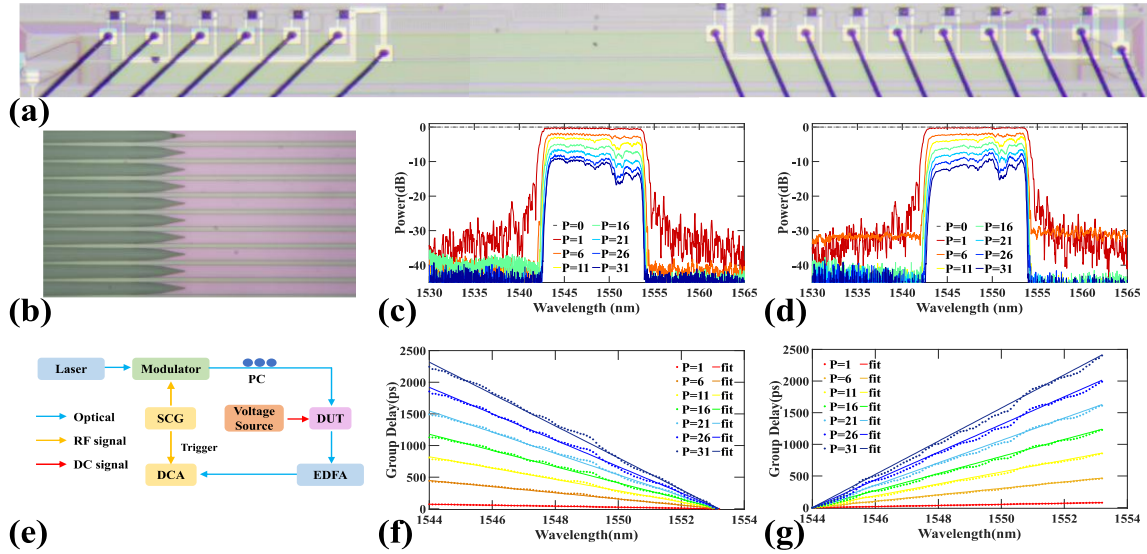


**Fig.2:** (a) Simulation results of GDR as apodization length ratio  $\beta$  varies. (b) Simulation spectral and group delay responses when selecting a single CMWG. (c, d) Simulated transmission spectra and (e, f) group delay profiles for the positive and negative dispersion of DTDC.

We characterized a five-stage DTDC configuration with a maximum capacity of 31 CMWG units. Figs. 2(c)–2(f) depict the systematic evolution of the transmission and group delay as more units are activated, spanning both positive and negative dispersion regimes. Under these conditions, the proposed DTDC achieves digital dispersion tuning ranging from  $-292.64 \text{ ps/nm}$  to  $310.3 \text{ ps/nm}$ , providing a total tuning span of  $602.94 \text{ ps/nm}$ . The discrete tuning steps are  $10.01 \text{ ps/nm}$  and  $-9.44 \text{ ps/nm}$  for the positive and negative dispersion regimes.

## 3. Fabrication and Characterization

The proposed DTDC was fabricated on a silicon-on-insulator (SOI) platform with a  $220\text{-nm}$  top



**Fig. 3:** (a-b) Microscopic images of (a) the fabricated DTDC; (b) slab-tapered transition area. (c-d) Measured transmission of the present DTDC operating with (c) positive and (d) negative dispersion when different numbers of CMWGs are involved. (e) Experimental setup for group delay measurement.; SCG: synthesized clock generator; DCA: digital communication analyzer; PC: polarization controller; EDFA: erbium-doped fiber amplifier. (f-g) Measured group delay of the present DTDC operating with (f) positive and (g) negative dispersion when different numbers of CMWGs are involved.

silicon layer and a 3- $\mu\text{m}$  buried oxide (BOX) layer via standard silicon photonics processes. Microscope images of the DTDC chip are shown in Fig. 3(a). A slab-tapered transition was implemented to facilitate a low-loss interface between the strip and rib waveguides, as shown in Fig. 3(b). A thin-film silica upper cladding was deposited, followed by the patterning of titanium nitride (TiN) heaters for the thermo-optic MZSs. The microheaters were connected to electrical pads through high-conductivity metal routing, which were subsequently wire-bonded to a printed circuit board (PCB).

To measure the transmission spectra of the devices on the chip, the setup with a broad-band amplified spontaneous emission light source and an optical spectrum analyzer was used. The experimental setup for characterizing the group delay and dispersion of the device is illustrated in Fig. 3(e). The continuous-wave light from a tunable laser source (TLS) was modulated by a 2–16 GHz Radio-frequency (RF) signal, provided by a synthesized clock generator (SCG), through an electro-optic intensity modulator. Polarization controllers (PCs) were employed to align the polarization state before the signal was coupled into the device under test (DUT). The output optical signal was then detected and analyzed by a digital communication analyzer (DCA). By scanning the wavelength and recording the temporal shifts of the time-domain waveforms, the relative group delay was accurately extracted. The thermo-optic MZSs were driven by direct current (DC) voltages supplied by a multichannel source, providing the active control necessary for switching and tuning during characterization.

Figures 3(c) and 3(d) illustrate the measured

transmission spectra of the DTDC under various configurations for both positive and negative dispersion regimes. Here, the number of active CMWGs is varied from 1 to 31 in steps of 5. The experimental results reveal that the average insertion loss per CMWG within a 11.68-nm 3dB bandwidth is 0.30 dB for the positive dispersion state and 0.31 dB for the negative dispersion state. At the maximum configuration, the propagation loss in the positive dispersion regime evolves from 9.37 dB at 1544nm to 13.4 dB at 1553.2nm. Conversely, for the negative dispersion regime, the loss scales from 12.79 dB at 1544 nm to 9.64 dB at 1553.2 nm. Within the spectral range of 1544–1553.2 nm, the dispersion is digitally tuned from -251.87 to 262.33 ps/nm with a discrete step size of 8.30 ps/nm per CMWG unit. The device exhibits low delay-related propagation losses, characterized at 1.67 dB/ns for positive dispersion and 1.40 dB/ns for negative dispersion.

#### 4. Conclusion

We have demonstrated a compact and low-loss digitally tunable dispersion controller based on rib-waveguide bidirectional chirped multimode waveguide gratings on a silicon-on-insulator platform. The experimentally realized device, occupying a footprint of  $4.85 \times 0.4 \text{ mm}^2$ , provides a digitally reconfigurable dispersion range from -251.87 to +262.33 ps/nm over the 1544–1553.2 nm wavelength band, while exhibiting low delay-dependent propagation losses of 1.67 dB/ns for positive dispersion and 1.40 dB/ns for negative dispersion.

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